

Nautilus — Literature Review

Software Group Project · Reporting 2

1. Personal Knowledge Management tools (Obsidian, Notion)

- **What it is:** Apps for keeping linked personal notes; Obsidian popularized local markdown files with `[[wikilinks]]` and a graph view.
- **Contribution:** Proved that linked, plain-text notes are a durable, portable way to store personal knowledge.
- **Limitation:** The notes are passive — the tool stores and displays them but cannot reason over them or answer questions.
- **What Nautilus takes:** The storage model, unchanged — an Obsidian-compatible vault of plain markdown is Nautilus's memory layer.

2. The 'Second Brain' methodology (Tiago Forte)

- **What it is:** A popular framework for personal notes: capture, organize, distill, express.
- **Contribution:** Showed that the bottleneck is human effort — most note systems die because organizing is manual work.
- **Limitation:** It is a habit method, not a technology — every step still depends on the person doing it consistently.
- **What Nautilus takes:** The workflow shape — but hands the 'organize' and 'distill' steps to scheduled AI tasks, keeping only effortless capture for the human.

3. Retrieval-Augmented Generation — RAG (Lewis et al., 2020)

- **What it is:** A technique where an LLM is given retrieved documents alongside the question, so its answer is grounded in those documents.
- **Contribution:** Reduces hallucination and lets a general model answer from private knowledge it was never trained on.
- **Limitation:** Research implementations assume embedding pipelines and vector databases — heavy machinery for a personal tool.
- **What Nautilus takes:** The pattern, simplified: keyword retrieval feeding the LLM, with sources cited in every answer. Embeddings remain an upgrade path, not a prerequisite.

4. AI features inside note tools (Notion AI, Smart Connections, Copilot plugins)

- **What it is:** Commercial and community attempts to add AI question-answering inside existing note apps.
- **Contribution:** Validated demand — users clearly want to talk to their notes.
- **Limitation:** Closed platforms or plugin-bound: the AI layer is rented, not owned; visuals are fixed; data often lives in someone else's cloud.
- **What Nautilus takes:** The goal, with ownership — the entire pipeline is the user's own code over their own files, with a purpose-built visual layer none of these tools offer.

5. Agentic automation (Claude Cowork, scheduled tasks)

- **What it is:** AI agents that operate on files and tools autonomously, on a schedule, in the cloud.
- **Contribution:** Makes 'a system that maintains itself' practical without writing automation code.
- **Limitation:** Requires trust and review — an agent that moves files must be supervised before it runs unattended.
- **What Nautilus takes:** Its Phase-4 features — auto-organizing, weekly digests, conflict checks — delivered as scheduled agent tasks writing into the vault.

6. The gap Nautilus fills — in one table

Existing approach	What it lacks — and Nautilus's answer
Note apps (Obsidian)	Store but cannot reason → Nautilus adds a grounded AI layer over the same files.
Generic AI chatbots	Reason but do not know the user → Nautilus grounds every answer in the user's own notes, with sources.
AI-note platforms	Owned by the vendor → Nautilus is user-owned, provider-agnostic, plain files, custom visuals.
Manual methodologies	Depend on discipline → Nautilus schedules the discipline: organizing, digesting and checking run themselves.